Mechanics of Flexible Structures & Soft Robots
MAE-263F
Fall 2026

Homework 3

Due Date: 11:59 PM, 11/04/2026

You should create a single GitHub repository for this class and share it with the instructor (`khalidjm@seas.ucla.edu`). All the homeworks, reports, and proposal should be uploaded to this repository. Do not create a separate repository for each assignment. Within your repository, create a separate folder for each assignment (e.g., `homework_1`, `homework_2`, `homework_3`, `homework_4`, `proposal`, `midterm_report`, and `final_report`).

Submission Instructions: This homework is graded entirely through the multiple-choice questions on **Gradescope**. In addition, submit the URL to your GitHub repository on BruinLearn. The `homework_3` folder of your repository should include the following items:

1. **Source code:** One main file named exactly `Homework3.py`, along with any additional files (e.g., functions) as necessary. The main file should require no more than a single command to run.
2. **README file:** A `README.md` file with clear instructions on how to run your code and a short description of each file in your repository.

Your code will not be graded, but it must be in your repository. If no code is submitted, the homework receives a score of 0, regardless of your Gradescope answers.

Introduction: From a Straight Beam to a Naturally Curved Rod

In class, we wrote a 3D Discrete Elastic Rods (DER) simulator (`main_discrete_elastic_rods.py` with the helper functions in `der_functions.py`, Chapters 7–8 of the notes). It simulates the naturally curved rod of the assignment on page 51 of the notes: one end is clamped and the rod deforms under gravity. This homework uses DER in two ways.

1. **Part 1: A straight, clamped–free rod under gravity.** You will simulate a straight rod clamped at one end (a cantilever) and compare its tip deflection with Euler–Bernoulli beam theory. The two agree when the load is small (small gravity or high Young's modulus). Beam theory fails when the deformation is large (high gravity or low Young's modulus); DER does not have this limitation.
2. **Part 2: The same code with natural curvature.** With minimal changes, you will use the same code for a naturally curved rod (the example in the notes) and look at twist and the role of the natural curvature.

Writing the code. You have access to `main_discrete_elastic_rods.py`. You may write your code from scratch (**recommended**) or modify the provided code once you understand all of its steps. The section "How to Modify the Provided Code" below lists every change needed.

Part 1: Straight Clamped–Free Rod Under Gravity

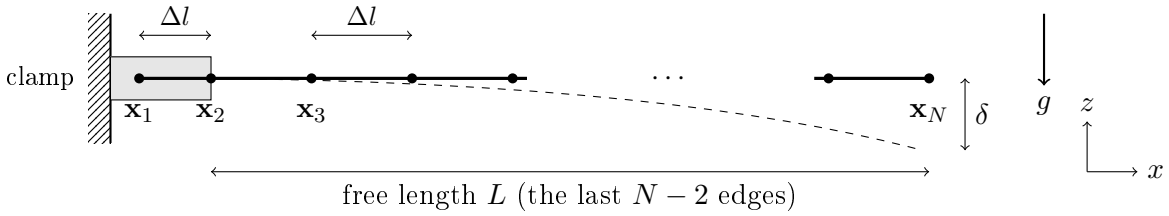


Figure 1: Straight clamped–free rod. The first two nodes ($\mathbf{x}_1$, $\mathbf{x}_2$) and the first twist angle θ^1 are fixed, so the first edge is held by the clamp. The free length L is the length of the last $N-2$ edges. The dashed curve sketches the deformed rod and the tip deflection δ.

Model. A straight rod with circular cross-section lies along the x-axis and is discretized into N nodes and $N-1$ edges of equal length Δl. Its DOF vector is $\mathbf{q} = [\mathbf{x}_1, \theta^1, \mathbf{x}_2, \theta^2, \ldots, \theta^{N-1}, \mathbf{x}_N]^T$ (size $4N-1$). The rod is **clamped–free**: the first two nodes and the first twist angle (the first seven DOFs) are fixed for all time. The first edge is held by the clamp, so the **free length** L of the rod is the length of the last $N-2$ edges:

$$\Delta l = \frac{L}{N-2}, \qquad \mathbf{x}_k = [\,(k-2)\,\Delta l,\ 0,\ 0\,], \quad k = 1, \ldots, N, \tag{1}$$

so that the clamped node $\mathbf{x}_2$ is at the origin and the tip $\mathbf{x}_N$ is at $x = L$. Gravity acts along $-z$. The rod starts straight and at rest; integrate in time until it reaches **steady state** (see below). The **tip deflection** is $\delta = -z_N$ at steady state, where z_N is the z-coordinate of the last node.

Default parameters.

Quantity	Value
Free length, cross-sectional radius	$L = 0.2\,\mathrm{m}$, $r_0 = 10^{-3}\,\mathrm{m}$
Young's modulus, Poisson's ratio	$E = 10\,\mathrm{MPa}$, $\nu = 0.5$ $(G = E/(2(1+\nu)))$
Stiffnesses	$EA = E\pi r_0^2$, $EI = E\pi r_0^4/4$, $GJ = G\pi r_0^4/2$
Density	$\rho = 1000\,\mathrm{kg/m^3}$
Gravity	$g = 9.81\,\mathrm{m/s^2}$ along $-z$ (unless a question says otherwise)
Discretization	$N = 52$ nodes (50 free edges), $\Delta t = 0.1\,\mathrm{s}$, total time $10\,\mathrm{s}$

Steady state. We only need the final (static) shape. Implicit Euler adds numerical damping, so the rod settles to its steady shape. With $\Delta t = 0.1\,\mathrm{s}$, the rod settles within a few seconds; with a smaller Δt (e.g., $0.01\,\mathrm{s}$, as in class), it oscillates for much longer and you would need a much longer (and slower) simulation. Run long enough that z_N stops changing: for example, check that z_N changes by less than 0.1% of $|z_N|$ over the last second of simulation. A total time of $10\,\mathrm{s}$ with $\Delta t = 0.1\,\mathrm{s}$ is enough for every question in this homework.

Watch the right quantity. Checking z_N works in Part 1 and in Question 9, where gravity moves the tip in z. It does *not* work in Question 11: there $g = 0$, the rod moves only in the x–y plane, and z_N stays zero even while the rod is still moving. In Question 11, check instead that the distance

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between the first and last nodes, $\|\mathbf{x}_N - \mathbf{x}_1\|$, has stopped changing (or, more generally, the whole DOF vector $\mathbf{q}$). In Question 10, also check that the twist angles θ^k have stopped changing.

Euler–Bernoulli beam theory. For a cantilever of length L under a uniform load q per unit length (here its own weight, $q = \rho A g$ with $A = \pi r_0^2$), linear beam theory gives the tip deflection

$$\delta_{\mathrm{EB}} = \frac{qL^4}{8EI}. \tag{2}$$

This formula assumes small slopes and that the shape does not change much from the straight configuration. Define the relative difference between the two predictions as

$$e = \frac{|\delta_{\mathrm{DER}} - \delta_{\mathrm{EB}}|}{\delta_{\mathrm{EB}}}. \tag{3}$$

Part 2: The Same Code with Natural Curvature

This is the assignment on page 51 of the notes. The centerline of the rod is now a circular arc of length 0.2 m (all $N-1$ edges) with natural radius $R_n = 0.02\,\mathrm{m}$. Because the edges are straight chords of this arc, the discrete rod is very slightly shorter (about 0.1997 m for $N = 50$); this does not affect the answers. Its nodes at $t = 0$ are

$$\mathbf{x}_k = [\, R_n \cos\left((k-1)\Delta\theta\right),\ R_n \sin\left((k-1)\Delta\theta\right),\ 0\,], \qquad \Delta\theta = \frac{0.2}{R_n}\,\frac{1}{N-1},$$

and the twist angles are $\theta^k = 0$ at $t = 0$. The rod lies in the x–y plane, and gravity ($g = 9.81\,\mathrm{m/s^2}$ along $-z$) pulls it out of that plane. As in Part 1, the first two nodes and the first twist angle are fixed. All other parameters (r_0, E, ν, ρ) are as in Part 1. Use $N = 50$, $\Delta t = 0.1\,\mathrm{s}$, and a total time of 10 s. This is exactly the setup of `main_discrete_elastic_rods.py` (except for Δt and the total time).

Natural curvature. In DER, the natural curvature $\bar{\boldsymbol{\kappa}}_k$ (`kappaBar` in the code) is computed from the initial shape. The bending energy is zero when the curvature equals the natural curvature, so the natural curvature sets the shape that the rod has when no load acts on it.

How to Modify the Provided Code

If you start from `main_discrete_elastic_rods.py`, these are the changes you need. Make sure you understand what each line of the code does before you change it.

1. **Make it a function.** Put the setup and the time-stepping loop inside a function, e.g., `run(N, L, E, g, natR, dt, totalTime)`, that returns the final DOF vector `q`. You will call it many times in Part 1.
2. **Turn off plotting and printing for speed.** Set `plotEvery` to a very large number (or remove `plotrod_simple` from the loop) and remove the `print` statements inside the time loop and in `objfun`. Plotting the rod at every step makes the code much slower.

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3. **Straight rod with free length L (Part 1).** The provided code places the straight rod (`natR` $= 0$) over all $N-1$ edges. Replace that branch with Eq. (1):

```
dl = L / (nv - 2)                # free length L = last nv-2 edges
for c in range(nv):
    nodes[c, 0] = (c - 1) * dl  # nodes[0] at x = -dl, nodes[1] at x = 0
```

   The total length of the rod is now $L + \Delta l$. Make sure the mass per edge is $\rho A \Delta l$ (in the provided code, `dm = totalM / ne`, so `totalM` must use the total length $(N-1)\Delta l$, not L).

4. **Parameters as inputs.** Make Young's modulus (`Y`), the gravitational acceleration (`g = np.array([0, 0, -gval])`), N, Δt (`dt`), and the total time (`totalTime`) inputs of your function.

5. **Boundary conditions.** The provided code already fixes the first seven DOFs (`fixedIndex = np.arange(0, 7)`), which is the clamped–free condition of this homework. Do not change it.

6. **Tip deflection and steady state.** The z-coordinate of the last node is `q[-1]`, and $\delta = -$`q[-1]`. Store it at every time step and check that it has stopped changing (see "Steady state" above). For Question 11, store the distance $\|\mathbf{x}_N - \mathbf{x}_1\|$ (`np.linalg.norm(q[-3:] - q[0:3])`) instead.

7. **Twist angles.** The twist angles are `theta = q[3::4]`. You will need them in Question 10.

8. **Natural curvature (Part 2).** Set `natR` $= 0.02$; the provided code then builds the curved rod of Part 2 with arc length `RodLength` $= 0.2\,\mathrm{m}$. For Question 11, add one line right after `kappaBar = getKappa(q0, m1, m2)`:

```
kappaBar = np.zeros_like(kappaBar)   # remove the natural curvature
```

Run time. With $N = 52$, $\Delta t = 0.1\,\mathrm{s}$, and a total time of $10\,\mathrm{s}$, one run takes well under a minute on a typical laptop (without plotting). The sweeps in Questions 5 and 6 need only a few runs each.

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Questions (answer on Gradescope)

Answer the following multiple-choice questions on Gradescope. Unless a question says otherwise, use the default parameters of Part 1. Choose the option **closest** to your result.

Part 1: Straight clamped–free rod

1. **Beam theory.** Using Eq. (2) with the free length $L = 0.2\,\mathrm{m}$, compute the Euler–Bernoulli tip deflection δ_{EB} for a **small gravity** $g = 0.05\,\mathrm{m/s^2}$ (all other parameters at their default values).

   (a) $0.5\,\mathrm{mm}$
   (b) $1.0\,\mathrm{mm}$
   (c) $4.0\,\mathrm{mm}$
   (d) $32\,\mathrm{mm}$

2. **Small load.** Simulate the rod with $g = 0.05\,\mathrm{m/s^2}$ until it reaches steady state. What is the DER tip deflection δ?

   (a) $1.0\,\mathrm{mm}$
   (b) $4.2\,\mathrm{mm}$
   (c) $16\,\mathrm{mm}$
   (d) $0.26\,\mathrm{mm}$

3. **Free length.** Why is the free length L the length of the last $N-2$ edges, and not the length of all $N-1$ edges?

   (a) The first two nodes are fixed, so the first edge cannot move; it is held by the clamp, and the rod leaves the clamp at $\mathbf{x}_2$.
   (b) The last edge does not carry any bending energy, so it should not be counted.
   (c) The first edge is needed only to store the first twist angle; it has no length.
   (d) The lumped mass of the first and last nodes is half of the mass of the other nodes, so one edge is lost.

4. **Mesh convergence.** Repeat Question 2 with $N = 22$, 42, and 82, keeping $L = 0.2\,\mathrm{m}$ fixed. Which statement best describes the DER tip deflection?

   (a) It is the same for every N, because the free length is the same.
   (b) It is smaller than δ_{EB} and moves further away from δ_{EB} as N increases.
   (c) It changes by more than a factor of 2 between $N = 22$ and $N = 82$.
   (d) It is slightly larger than δ_{EB} and approaches δ_{EB} as N increases; the difference roughly halves each time the number of free edges doubles.

5. **Stiffer rods.** Use the full gravity $g = 9.81\,\mathrm{m/s^2}$ and simulate the rod for $E = 10$, 30, 100, 300, and $1000\,\mathrm{MPa}$. What is the *smallest* of these values of E for which the relative difference e (Eq. (3)) is below 20%?

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(a) 10 MPa

(b) 30 MPa

(c) 100 MPa

(d) 1000 MPa

6. **Heavier load.** Use $E = 10\,\text{MPa}$ and simulate the rod for $g = 0.1$, 0.3, 1, 3, and $9.81\,\text{m/s}^2$. What is the *smallest* of these values of g for which the relative difference e is above 20%?

   (a) $0.1\,\text{m/s}^2$

   (b) $0.3\,\text{m/s}^2$

   (c) $1\,\text{m/s}^2$

   (d) $3\,\text{m/s}^2$

7. **Low stiffness.** Use $E = 10\,\text{MPa}$ and $g = 9.81\,\text{m/s}^2$ (the default parameters). Simulate long enough for the rod to reach steady state. What is the DER tip deflection δ?

   (a) 0.02 m

   (b) 0.18 m

   (c) 0.06 m

   (d) 0.78 m

8. **Why they differ.** In Question 7, the Euler–Bernoulli prediction is very different from the DER result. What is the main reason?

   (a) Euler–Bernoulli theory is linear: it assumes small slopes and a shape that stays close to the straight configuration. DER makes no such assumption.

   (b) Implicit Euler adds artificial damping, which reduces the DER deflection at large loads.

   (c) The rod stretches a lot at large loads, and Euler–Bernoulli theory ignores stretching.

   (d) DER includes the mass (inertia) of the rod, while Euler–Bernoulli theory does not.

Part 2: Naturally curved rod

Use the setup of Part 2: $N = 50$, $\Delta t = 0.1\,\text{s}$, total time 10 s, $g = 9.81\,\text{m/s}^2$, unless stated otherwise.

9. **Curved rod.** What is the z-coordinate of the last node at steady state?

   (a) $-0.01\,\text{m}$

   (b) $-0.18\,\text{m}$

   (c) $-0.04\,\text{m}$

   (d) $-0.10\,\text{m}$

10. **Twist.** Compare the twist angles θ^k (`q[3::4]`) of the straight rod in Question 7 with those of the curved rod in Question 9. Which statement is true?

    (a) Both rods develop nonzero twist angles, because gravity always twists a rod.

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(b) The twist angles of the straight rod stay zero, because it bends only in its own plane; the curved rod develops nonzero twist angles, because gravity pulls it out of its plane.

(c) The twist angles of both rods stay zero, because the first twist angle is fixed.

(d) The straight rod develops nonzero twist angles, but the curved rod does not, because its natural curvature prevents twisting.

11. **No natural curvature.** Set the natural curvature to zero (add the line given in "How to Modify the Provided Code") and set $g = 0$. Start from the same curved initial shape and simulate until steady state. Which statement best describes the result?

(a) The rod keeps its initial curved shape, because there is no load.

(b) The rod curls up more tightly, because its curvature increases.

(c) The simulation fails, because Newton's method cannot converge when the natural curvature is zero.

(d) The rod opens up into a nearly straight rod: the distance between its first and last nodes becomes close to 0.2 m.

Appendix: Useful Relations

- Cross-sectional area and moments: $A = \pi r_0^2,\ \ I = \pi r_0^4/4,\ \ J = \pi r_0^4/2$.
- Weight per unit length: $q = \rho A g$.
- Euler–Bernoulli cantilever under its own weight (tip deflection): $\delta_{\mathrm{EB}} = qL^4/(8EI)$. Note that δ_{EB} grows with L^4; the free length L must be used.
- For a rod with circular cross-section, $qL^4/(8EI) = \rho g L^4/(2Er_0^2)$.

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